

- 6 (a) *either* $P = VI$ and $V = IR$ *or* $P = V^2 / R$ C1
 resistance = $38.4 \, \Omega$ A1 [2]
- (b) zero B1
 1.5 kW B1
 3.0 kW B1
 0.75 kW B1
 2.25 kW B1 [5]

- 7 (a) α particle: *either* helium nucleus *or* contains 2 protons + 2 neutrons